

Rethinking case attraction on Ancient Greek infinitive clauses

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1 Introduction

Good morning. Nominal predicates and adjuncts modifying subjects of complement infinitive clauses in Ancient Greek (AG) usually agree in case with their domain local subject, which has morphological accusative case assigned in the *Accusativus cum Infinitivo* (*AcI*) construction. When the subject of the infinitive is not overtly expressed but the main clause has a dative or genitive object which shares the reference of the infinitive subject, these predicates or adjuncts may either occur in the accusative or “be attracted” to the case of the main clause’s object.

- (1) a. συμβουλεύει τῷ Ξενοφῶντι ἐλθόντα εἰς Δελφοὺς ἀνακοινῶσαι τῷ θεῷ περὶ τῆς πορείας.
He advises Xenophon to go to Delphi and ask the god about the travel. (Xen. *Anab.* 3.1.5)
- b. ἀφῆκε μοι ἐλθόντι πρὸς ὑμᾶς λέγειν τὰληθῆ.
He allowed me to go and speak the truth in front of you. (Xen. *Hell.* 6.1.13)

The literature on the subject, on the more philological side, from Port-Royal (Lancelot 1658, 1709) up to the Cambridge Grammar (Boas et al. 2019), have focused on listing the contexts in which attraction may occur, whereas the transformational and minimalist approaches of Lakoff (1970), Andrews (1971), Quicoli (1982), Spyropoulos (2005), Tantalou (2003), Sevdali (2007, 2013) have

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mainly occupied themselves with proposing the structure that would allow the phenomenon to occur. What factors influence the decision between either resolution in (1) remains, however, largely unknown.

[Self-deprecatingly] I myself have, in the last edition of this same event, tried to present the distributional patterns that make the attraction, (1-b), more likely, showing that there is strong correlation between attraction and the classes of the matrix and infinitive verbs and the relative distance and position of the controller and target of agreement.

Furthermore, I argued that these factors were *proxies* to unobserved values, namely semantic and pragmatic features of the discourse setting, and that even in the absence of seen contrasting factors, precisely the case in the pair above (1), the unseen discourse factors would be still causing one resolution to be at the very least more likely.

[Introductingly] I wasn't, however, quite convinced about neither the methods, the data, and, consequently, the quality of the results at which I had arrived. The main reason for this distrust is the fact that there could be interaction between the factors analysed which the methods employed were not well designed to account for, at least not with the amount of data available. To solve these shortcomings, we must take some steps back and build a model for what is the linguistic nature of case attraction

The aim of this paper is to walk through the assumptions present in my own and other's assessments of the phenomenon, as I believe most of them are in themselves right, build from these assumptions a causal system and show how much of this system is left unexplained with what we do know and, more importantly, I believe, what we **can** know and measure about ancient languages in general.

2 The data

Before stepping into the framework and analysis proper, a brief word on the empirical data supporting this research and any conclusion coming from it. The dataset is composed of 343 sentences drawn from the Attic literary *corpus* and Herodotus. The sentences were extracted from the Diorisis Corpus (Vatri and McGillivray 2018, 2020) with help of the project's querying tool, Diorisis Search (Vatri 2020-2022). Although comprehensive, the selection is by no means exhaustive.

Each sentence has been annotated for values of interest, including authorship, literary genre, dialect as well as class of matrix and infinitive verb, class of

controller and target, distance between constituents, etc. Parts of the annotation processes were conducted programmatically and later checked manually.

Every model used in this work was processed using R (R Core Team 2013), the packages tidyverse (Wickham et al. 2019), dagitty (Textor et al. 2016), ggdiag (Barrett 2025), and rethinking (McElreath 2020), as well as the software Stan (Stan Development Team 2025a,b).

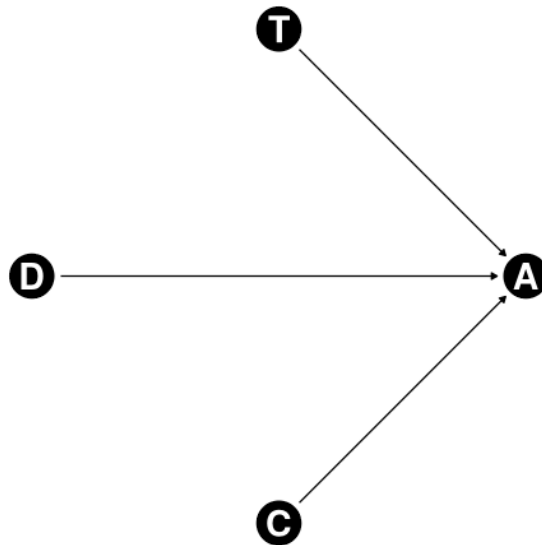
3 Nature of attraction

In the most abstract way possible, case attraction in infinitive clauses, exemplified by (1-b), is when two constituents, a controller and a target, produced in different domains of a sentence, one of the domains encompassing the other, covary in an specific feature, namely [*Case*]. This satisfies the definition of agreement in Corbett’s typological assessment of the phenomenon across languages (2006, pp. 4–7).

The lack of covariation of the [*Case*] feature, seen in (1-a), is generally understood as result of the application of **default** case concord rules of Ancient Greek. This assumes that the subject of infinitive clauses in Greek is always marked as accusative, even when not overtly expressed, and that convert constituents are also marked for [*Case*] and its case is also available for nominal concord.

The facts that the target has a choice of controller (either the overt oblique or de convert accusative) for agreement, that [*Case*] is a non-lexical features, that the agreement is optional, and that the domain is non-local favours the interpretation that the agreement type of case attraction is less canonical than the general agreement type of Ancient Greek, following the criteria for canonical agreement in the same work of Corbett (2006, pp. 8ff.).

If this assessment is sound, it is expected that attraction and non-attraction behave as competing agreement resolutions, the former being less canonical and thus more likely to be conditioned. Conditions concerning the semantic and pragmatic properties of the controller, its precedence in relation to the target and the class of predication (i.e.: class of the domain) have been identified by Corbett. Although properties of the target were not shown to be favouring any resolution in agreement mismatches in Corbett’s survey, case attraction might be one example of it, and I will hypothesize that properties of targets are competing causes of non-canonical resolutions. A model of agreement in this framework may be represented as a direct acyclic graph, in which Attraction ($\simeq$ Agreement) is caused by features of the **C**ontroller, **D**omain and **T**arget:



These Controller, Domain and Target nodes represent clusters of linguistic factors, some of which we may observe directly (part of speech of the target), some might be implied (animacy) and some may remain unobserved (discourse function, as Ancient Greek does not overtly mark focuses or topics in the morphosyntax, at least not dependably).

The three factors previously shown to correlate to case attraction represent each of these nodes:

- Copular infinitives imply a different domain of agreement, as they select different predication classes;
- Main verbs with semantic modality imply a different type of controller, as they have higher subjecthood;
- Word order and distance imply different status for the target, as targets more to the left of the sentence are both structurally closer to the controller and have a higher communicative function.

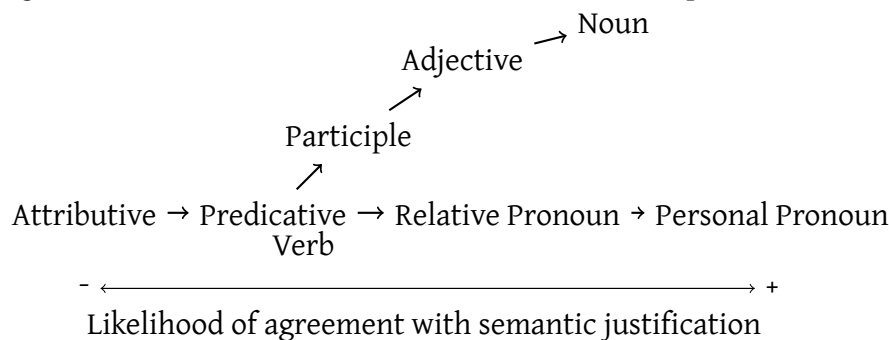
4 The domain of attraction

Firstly, Kühner and Gerth (1898) and others have associated case attraction with sentences in which the infinitival clause is a copular infinitive, *einai* or *genesthai*. Should the class of the infinitive favour either resolution of agreement? Probably not directly, as there is no reason linguistically it would. Copular infinitives, however, have a double effect on the agreement “causes”:

1. they filter the possible relations between controller and target, as those are now more likely in primary predication, rather than secondary or appositive relation;
2. this then delimits the option of target, making it more likely that they are nouns or adjectives, rather than participles.

These are known to affect agreement resolutions, and a couple of Hierarchies have been proposed to predict when the less canonical resolution (also called semantic agreement) is more likely to take place. The Agreement Hierarchy (Corbett 1979) and Predicate Hierarchy (Comrie 1975) have also been combined into the Agreement and Predicate Hierarchy shown in (1):

(1) Agreement and Predicate Hierarchies (Corbett 2006, p. 233):



Adding this to our causal model, we may say that **Attraction** is caused both by whether or not the relation between the controller and target is copular, and by the Part of Speech of the target, which is itself “caused” by the type of relation (copulas select more nominal predicates on average), and other unobserved properties of the **Domain** may also cause Attraction.

By applying the do-calculus (Pearl 2009), we learn that:

- we can not estimate neither the total or direct effect of the infinitive verb being copular, as we can not close the non-causal path $C \leftarrow D \rightarrow A$;
- the direct and total effect of the part of speech class requires us to adjust the correlation by C , so that we block the non-causal path $P \leftarrow C \leftarrow D \rightarrow A$.

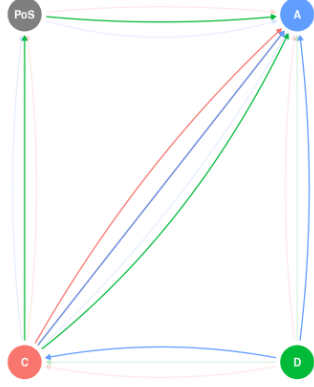


Figure 1: Open causal paths between C and A . The green node is unobserved. The blue path is non-causal and can not be closed.

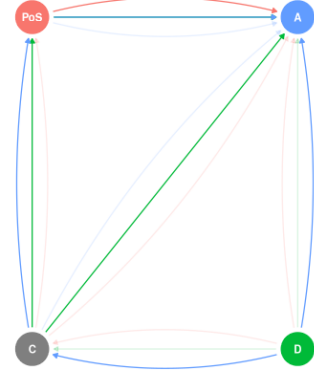


Figure 2: Open causal paths between PoS and A . The green node is unobserved. The blue path is non-causal and must be closed by adjusting for C , which also closes the green path.

To estimate the effect of PoS over Attraction, we can use the generalized linear model below. We can control for Author, as there is no reason to believe that any author from the corpus prefers copulas or any type of target. As it is usually agreed that typological hierarchies increase their effect monotonically by each of their nodes, but we do not know if their effect should be linear or not, a Dirichlet prior is suitable to express the effect of Part of Speech. We coarse the base likelihood of case attraction to be negative as we assume that it is the non-canonical agreement resolution, meaning it should be less likely to occur.

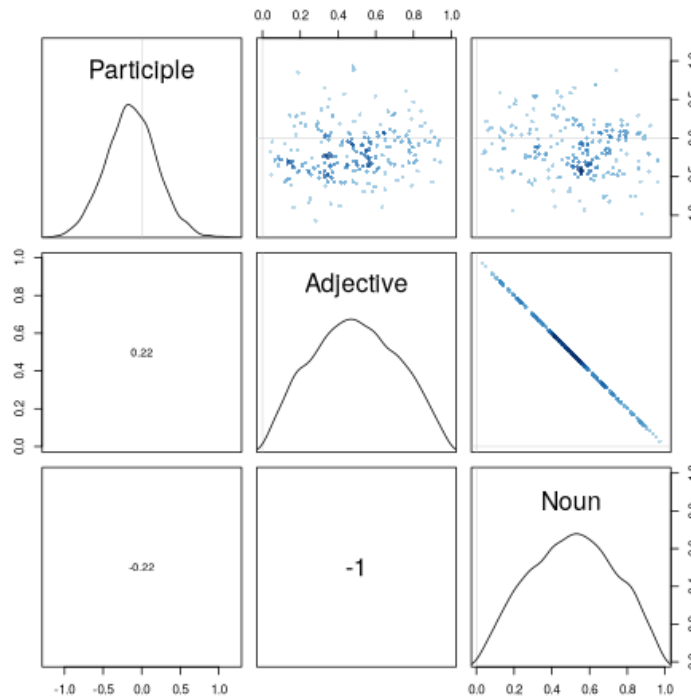
$$\begin{aligned}
A &\sim \text{Bernoulli}(p_i) \\
\text{logit}(p_i) &= \alpha + \beta_{PoS} \sum_{j=0}^{PoS_i-1} \delta_j + \beta_C C_i + \beta_{Auth_i} \\
-\alpha &\sim \text{LogNormal}(0, 1) \\
\beta_{PoS} &\sim \text{Normal}(0, 1) \\
\delta &\sim \text{Dirichlet}(2) \\
\beta_C &\sim \text{Normal}(0, 1) \\
\beta_{Auth_j} &\sim \text{Normal}(\bar{\beta}_{Auth}, \sigma_{Author}) \\
\bar{\beta}_{Auth} &\sim \text{Normal}(0, 1) \\
\sigma_{Auth} &\sim \text{Exponential}(1)
\end{aligned}$$

The priors are fairly flat, as there is no reason to believe we have weak or strong relationships in agreement phenomena. β_{PoS} is the general effect of the part of speech over the likelihood of Attraction, it is slightly negative, and δ is a vector with values denoting how much going up from Participle to Adjective to Noun increases β_{PoS} . Participles do not change the effect, Adjectives add δ_1 and Nouns add $\delta_1 + \delta_2$, meaning that the effect is expected to be cumulative.

The estimated effects are described in the following table and plot.

Effect	Mean	SD	5.5%	94.5%
β_{PoS}	-0.13	0.33	-0.66	0.40
δ_{Adj}	0.49	0.22	0.14	0.85
δ_{Noun}	0.51	0.22	0.15	0.86

PoS	Cumulative effect	Mean	SD	5.5%	94.5%
Participle	β_{PoS}	-0.13	0.33	-0.66	0.40
Adjective	$\beta_{PoS} + \delta_{Adj}$	0.36	0.44	-0.33	1.06
Noun	$\beta_{PoS} + \delta_{Adj} + \delta_{Noun}$	0.87	0.33	0.32	1.38



To put it into words, the data favours the idea that the Agreement and Predication Hierarchy holds for case attraction in Ancient Greek case attraction: as the target becomes more “nominal”, the likelihood of attraction increases from slightly negative up to highly positive. Notice that the model assumes that the effect of adjectives and nouns are negatively correlated, meaning that it does not discount the possibility that the bulk of the effect is in the passage from Participles to Adjectives, and going further up to a Noun makes little difference.

5 The controller of attraction

Another association is that between case attraction and impersonal verbs. I have, however, shown that this association does not exist with every single impersonal verb, and that the verb *δοκέω* almost never heads a sentence with case attraction. The impersonal verbs which are actually associated with case attraction are those carrying modal semantics, such as *ἐξαρκεῖ* and *ἔξεστί*. Is it likely that the nature of the matrix verb directly affects the choice of agreement? I don’t think so, but it may indirectly do, by defining which features the controller of agreement, the oblique object, has. These verbs have very little semantic information and the theta role and grammatical function of their oblique object seems to depend on the infinitive verb. We may say that *σοι* in (1), for example, is the subject of the sentence (a quirk subject of sorts) and has the thematic role of agent, much in contrast if the sentence was constructed with a verb of order or advice like *συμβουλεύω*.

- (1) ἄλλ’ ἐξαρκῇ σοι τὰ ἔργα αὐτῶν ὁρῶντι σέβεσθαι καὶ τιμᾶν τοὺς θεούς.
But it is enough for you to look at these deeds and respect the gods. (Xen. *Mem.* 4.3.13)

6 The target of attraction

Finally, it has been long been proposed that case agreement is more likely when controller and target are very close to each other, and the likelihood diminishes as it grows. I have previously showed that, although very much associated, long distance between controller and target is not a dependable predictor for lack of attraction, but that precedence of the target in relation to the controller seems to demand case attraction, such as in (1).

- (1) ὥς ἀγαθοῖς τε ὑμῖν προσήκει εἶναι.
That you may be good people. (Xen. *Anab.* 3.2.11)

As with the previous factors, we must ask: are distance and precedence directly causing case attraction, or are they indirect causes? Longer distances have been shown to favour non-canonical agreement resolutions in other languages, such as with personal pronouns agreeing with committee nouns in English, which are more likely to be plural the further away from their controller. This is the very opposite of what is being proposed for case attraction: the canonical agreement (the local one) is the one becoming more likely as farther away the target is from the controller.

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